Vectors and Geometry

Dot and Cross Products

$$a \cdot b = |a||b| \cos(\theta)$$

$$|a \times b| = |a||b| \sin(\theta)$$

$$a \times b = \det \begin{pmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix}$$

So, two vectors are orthogonal $\Longleftrightarrow a \cdot b = 0$.

Direction cosigns are defined as $\cos \alpha = \frac{a \cdot i}{|a| \cdot |i|}$ and with b and c is typically notated as β and γ

Volumes

The volume of the parallelpiped with vectors a, b, c is $a \cdot (b \times c)$

Lines and Planes

The line is $r = r_0 + tv$ or equivilently $\frac{x - x_0}{v_1} = \frac{y - y_0}{v_2} = \frac{z - z_0}{v_3}$.

The plane is given by $n \cdot (r - r_0) = 0$ or wlog $n_1 x + n_2 y + n_3 z + d = 0$ for some d.

The distance between two planes is 0 unless they're parallel ($n_1 = n_2$), in which case it's $\frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$.

The distance between a point and a plane is $\frac{|a x_0 + b y_0 + c z_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$

The distance between a line and a plane is 0 unless they're parallel ($v \cdot n = 0$) in which case all points are equidistant to the plane, so pick any point and use the point to plane formula.

The distance between two lines that intersect is 0, the distance between two lines $r_1 = a + \lambda d_1$ and $r_2 = b + \mu d_2$ that are parallel is always equal, so pick a point on one and use the point to line. Otherwise (they're skew), use $\frac{|(b - a) \cdot (d_1 \times d_2)|}{|d_1 \times d_2|}$

The distance between a point and a line is $\frac{|(b - a) \times d|}{|d|}$

Tangent Vectors

$r'(a)$ is the tangent vector of the curve at point a.

Then the unit tangent vector is given by $T(t) = \frac{r'(t)}{|r'(t)|}$

Then the curvature is $\kappa = \left|\frac{\mathrm{d}T}{\mathrm{d}s}\right| = \left|\frac{\mathrm{d}T}{\mathrm{d}t} \frac{\mathrm{d}t}{\mathrm{d}s}\right|$. So, $\kappa = \frac{|T'(t)|}{|s'(t)|} = \frac{|(T'(t))|}{|r'(t)|}$. Further, $\kappa = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}$.

The principle unit normal vector is given by $N(t) = \frac{T'(t)}{|T'(t)|}$

The binormal vector is given by $B(t) = T(t) \times N(t)$.

Also, $N(t) = B(t) \times T(t)$ and $T(t) = N(t) \times B(t)$, and $\{T, B, N\}$ form a right handed basis of unit vectors for $\mathbb{R}^3$

Partial differentiation

Limits of functions of several variables

If a function is of multiple variables, e.g. $f(x_0, x_1, ..., x_i)$ we can take a limit across multiple of them $\lim_{(x_0, x_1, ..., x_i) \to (a_0, a_1, ..., a_i)} f = L$, which is only defined if *every* path to the point a approaches L.

For instance, we can check the limit from the x_0-axis by holding all other variables constant. Or along a line by parametrizing all other variables with respect to one variable.

Partial derivitives

Remember that the derivitive of $g(x)$ is defined as a limit with respect to x. If we apply the same to f we can then take that limit as it approaches from a single variable. This is called a partial derivitive, notated as $\frac{\mathrm{d}f}{\mathrm{d}x_0}$ or f_{x_0}. Note that $f_{xy} = f_{yx}$ (derivitives commute).

We can take the directional derivitive by applying this limit along the line defined by some vector u. This reduces to $D_u f(x, y) = u_0 f_x(x, y) + u_1 f_y(x, y)$. This is maximized when u is the gradient vector.

We define the gradient, $\nabla f(x, y) = \begin{pmatrix} f_x & f_y \end{pmatrix}^T$

General chain rule

Suppose z is a differentiable function of $x_1, x_2, ..., x_n$ and each of x_j is a differentiable function of $t_1, t_2, ..., t_m$. Then,

$$\frac{\mathrm{d}z}{\mathrm{d}t_i} = \frac{\mathrm{d}z}{\mathrm{d}\ \text{x\_1}} \frac{\mathrm{d}\ \text{x\_1}}{\mathrm{d}t_i} + \frac{\mathrm{d}z}{\mathrm{d}\ \text{x\_2}} \frac{\mathrm{d}\ \text{x\_2}}{\mathrm{d}t_i} + ... + \frac{\mathrm{d}z}{\mathrm{d}\ \text{x\_n}} \frac{\mathrm{d}\ \text{x\_n}}{\mathrm{d}t_i}$$

For each $i = 1, 2, ..., m$.

Implicit differentiation

Implicit Function Theorem: If $F(x, y, z)$ is defined within a sphere containing (a, b, c) where $F(a, b, c) = 0$, $F_z(a, b, c) \neq 0$ and F_x, F_y, F_z are continuous inside the sphere, then $F(x, y, z) = 0$ defines z as a differentiable function of x and y near the point (a, b, c) with partial derivitives below.

$$\frac{\mathrm{d}z}{\mathrm{d}x} = -\left(\frac{\frac{\mathrm{d}F}{\mathrm{d}\ \text{x}}}{\frac{\mathrm{d}F}{\mathrm{d}z}}\right) \qquad \frac{\mathrm{d}z}{\mathrm{d}y} = -\left(\frac{\frac{\mathrm{d}F}{\mathrm{d}\ \text{y}}}{\frac{\mathrm{d}F}{\mathrm{d}z}}\right)$$

Differentials

$$dz = f_x(x, y)dx + f_y(x, y)dy = \frac{\mathrm{d}z}{\mathrm{d}x}dx + \frac{\mathrm{d}z}{\mathrm{d}y}dy$$

Tangent planes to level surfaces

The tangent plane to the level surface $F(x, y, z) = k$ as the plane that passes through (x_0, y_0, z_0) and has normal vector $\nabla F(x_0, y_0, z_0)$ (provided $\nabla F(x_0, y_0, z_0) \neq 0$).

Second derivitives test

$$D(a, b) = \left|\begin{pmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{pmatrix}\right|$$

$$D > 0 \text{ and } f_{xx} > 0 \Rightarrow \text{local min}$$

$$D > 0 \text{ and } f_{xx} < 0 \Rightarrow \text{local max}$$

$$D < 0 \Rightarrow \text{saddle point where the graph crosses a tangent line}$$

$$D = 0 \Rightarrow \text{inconclusive}$$

General form is called the Hessian Matrix.

Lagrange multipliers

Say we have $L(x, y, \lambda) = f(x, y) - \lambda g(x, y)$, where f is what we want to maximize and subject to constraints $g(x, y) = 0$. Then any critical point of L will be a critical point of f subject to g and λ is called a lagrange multiplier.

Distance between a surface and a point

For a surface $F(x, y, z) = 0$, minimize the distance between P and Q, where Q is some point on the plane and P is the (usually known) point we're trying to minimize the distance to, subject to the constraint of the surface given (often using Lagrange multipliers).

Iterated Integrals

Trig Identities

$$\sin^2\theta = \frac{1 - \cos(2\theta)}{2} \qquad \cos^2\theta = \frac{1 + \cos(2\theta)}{2}$$

$$\sin(2\theta) = 2 \sin\theta \cos\theta \qquad \cos(2\theta) = \cos^2\theta - \sin^2\theta$$

General Jacobian

$$J_f = \begin{pmatrix} \frac{\mathrm{d}f}{\mathrm{d}x_1} & \frac{\mathrm{d}f}{\mathrm{d}x_2} & ... & \frac{\mathrm{d}f}{\mathrm{d}x_n} \end{pmatrix} = \begin{pmatrix} \frac{\mathrm{d}f_1}{\mathrm{d}x_1} & ... & \frac{\mathrm{d}f_1}{\mathrm{d}x_n} \\ ... & ... & ... \\ \frac{\mathrm{d}f_m}{\mathrm{d}x_1} & ... & \frac{\mathrm{d}f_m}{\mathrm{d}x_n} \end{pmatrix}$$

Say $(x, y) \to (u, v)$. First, solve for x, y in terms of u, v, then use that as f in the J_f calculation, and take the determinate. That's what you'll multiply by in the integral.

Polar Coordinates

$$y = r \sin\theta \qquad x = r \cos\theta \qquad dA = r dr d\theta$$

Cylindrical Coordinates

$$x = \rho \cos\varphi \qquad y = \rho \sin\varphi \qquad z = z \qquad dV = \rho d\rho d\varphi dz$$

Spherical Coordinates

$$x = r \sin\theta \cos\varphi \qquad y = r \sin\theta \sin\varphi \qquad z = r \cos\theta$$

$$dV = r^2 \sin\varphi d\theta d\varphi dr$$

Vector Calculus

Vector Fields

For a scalar function f in $\mathbb{R}^3$, the gradient is $\nabla f = (f_x, f_y, f_z)$.

Conservative Vector Fields

A vector field is conservative $\iff F = \nabla f$ for some f. f is then the potential function for F.

A vector field F is independent of path iff it is conservative.

Path Integrals

$$\int_C f(x,y) ds = \int_a^b f(r(t))|r'(t)|dt$$

$$\int_C f(x,y) \cdot dr = \int_a^b (f(r(t)) \cdot r'(t))dt$$

Fundamental Theorem for Line Integrals

$$\int_C \nabla f \cdot dr = f(r(b)) - f(r(a))$$

If f is a smooth differentiable function on C and C is smooth.

Green's Theorem

A **simple-curve** is one that does not intersect itself anywhere between its start and end. A **simply connected region** in the plane is a connected region D such that every simply closed curve encloses only points in D.

If C is a positively oriented, piecewise smooth, simple closed curve in the xy plane and D is the region enclosed by C and if P and Q have continuous partial derivatives on an open region that contains D.

$$\int_C P dx + Q dy = \iint_D \left(\frac{\mathrm{d}Q}{\mathrm{d}x} - \frac{\mathrm{d}P}{\mathrm{d}y}\right) dA$$

Traverse the curve in the path integral in a counterclockwise direction.

Curl and Divergence

Let $F = (P, Q, R)$,

$$\nabla = \left(\frac{\mathrm{d}}{\mathrm{d}x}, \frac{\mathrm{d}}{\mathrm{d}y}, \frac{\mathrm{d}}{\mathrm{d}z}\right)$$

$$\text{curl } F = \nabla \times F = \begin{pmatrix} R_y - Q_z \\ R_x - P_z \\ Q_x - P_y \end{pmatrix} \qquad \text{div } F = \nabla \cdot F$$

An **irrotational** field has curl $F = (0,0,0)$ everywhere in the domain.

A **conservative** field is irrotational and has a simply connected (no holes) domain.

Surface Integrals

We parametrize the surface S as $\underline{r}(u,v)$

$$\iint_S f(x,y,z) dS = \iint_D f(r(u,v)) |r_u \times r_v| dA$$

If $z = g(x,y)$ and D is its projection on the xy plane under S

$$\iint_S f(x,y,z) dS = \iint_D f(x,y,g(x,x)) \sqrt{1 + \frac{\mathrm{d}z}{\mathrm{d}x}^2 + \frac{\mathrm{d}z}{\mathrm{d}y}^2} dA$$

If $F(x,y,z) = c$, then we can project onto the xy plane, and get

$$\iint_S f(x,y,z) dS = \iint_D f(x,y,z) \frac{|\nabla F|}{|\nabla F \cdot p|} dA \qquad p = \hat{k}$$

Common Parametrizations

Surface	Parametrization
$z = f(x,y)$	$\boldsymbol{r}(x,y) = \langle x, y, f(x,y) \rangle$
$ax + by + cz = d$	Solve for one variable, e.g. $z = \frac{d-ax-by}{c}$, then $\boldsymbol{r}(x,y) = \langle x, y, \frac{d-ax-by}{c} \rangle$
$x^2 + y^2 = R^2$	$\boldsymbol{r}(\theta, z) = \langle R\cos\theta, R\sin\theta, z \rangle$
Cylinder about the x-axis	$\boldsymbol{r}(x, \theta) = \langle x, R\cos\theta, R\sin\theta \rangle$
Sphere of radius R	$\boldsymbol{r}(\varphi, \theta) = \langle R\sin\varphi\cos\theta, R\sin\varphi\sin\theta, R\cos\varphi \rangle$
Cone $z = r$	$\boldsymbol{r}(r, \theta) = \langle r\cos\theta, r\sin\theta, r \rangle$
Surface of revolution	Replace the circular coordinates with $r\cos\theta$ and $r\sin\theta$.

Surface Integrals of Vector fields

$\hat{n}$ determines the **orientation** of the surface (the direction its pointing in is the positive side). A smooth surface is **orientable** if there exists such an $\hat{n}$ that varies continuously on S and is normal everywhere on S. A piecewise smooth surface is orientable if whenever 2 component surfaces join at a boundary curve C, they induce opposite orientations along C.

$$\hat{n} = \frac{r_u \times r_v}{|r_u \times r_v|}$$

$$\iint_S F \cdot d\underline{S} = \iint_S F \cdot \hat{n} dS$$

Stokes Theorem

Let S be an oriented piecewise smooth surface bounded by a simple closed piecewise smooth boundary curve C with orientation inherited from S. Let F be a vector field with continuous partial derivitives on an open region that contains S, then,

$$\int_C F \cdot dr = \iint_S \text{curl } F \cdot dS = \iint_S (\nabla \times F) \cdot \hat{n} dS$$

If the surface S lies in the xy plane with upward orientation, then stokes theorem gives

$$\int_C F \cdot dr = \iint_S F \cdot k dA$$

The Divergence Theorem

Let E be a simple 3-dimensional region whose boundary surface S has positive (i.e., outward) orientation. Let F be a vector field whose component functions have continuous partial derivatives on an open region that contains E. Then,

$$\iint_S F \cdot ds = \iiint_E \text{div } F dV$$

Classification of DEs

An **ODE** includes derivatives with one variable, while a **PDE** includes more than one variable.

An n-th order **linear** ODE has the general form:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_0 y(x) = f(x)$$

Where $a_n, a_{n-1}, \ldots, f(x)$ are functions of x. If $f(x) = 0$ then the ODE is **homogenous**, else the ODE is **non-homogenous**.

Principle of Superstition

If $y = y_1(x)$ and $y = y_2(x)$ are particular solutions to a linear homogenous ODE, then so is $y = Ay_1(x) + By_2(x)$ for any constants A, B.

First-Order ODEs

Considering the general first order ODE $\frac{\mathrm{d}y}{\mathrm{d}t} = f(t,y)$, there's no single way of solving with elementary functions, instead we break into subclasses.

Linear Equations

An equation of the form,

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

Can be solved by multiplying both sides by the integrating factor $e^{\int P(x)dx}$, and yields,

$$y = \frac{1}{e^{\int P(x)dx}}\left[\int e^{\int P(x)dx}Q(x)dx + C\right]$$

Seperable Equations

An equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)h(x)$$

Can be seperated and integrated, yielding

$$\int \frac{1}{h(y)}dy = \int g(x)dx$$

Homogenous Equations

The ODE

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x,y) \text{ is homogenous if } f(x,y) = g\left(\frac{y}{x}\right)$$

Note: not like homogenous used in the classifications before!

These can be solved with the change of variables

$$v(x) = \frac{y}{x}$$

Exact Equations

The ODE

$$M(x,y) + N(x,y)\frac{\mathrm{d}y}{\mathrm{d}x} = 0$$

is **exact** if there exists ψ such that

$$\frac{\mathrm{d}\psi}{\mathrm{d}x} = M(x,y) \text{ and } \frac{\mathrm{d}\psi}{\mathrm{d}y} = N(x,y)$$

And the solution is

$$\psi(x,y) = C$$

You can find ψ by integrating M wrt x and N wrt y and finding a solution which works for both.

Integrating Factors

When the ODE is not exact, we may be able to convert it to one by multiplying by the integrating factor $\mu(x,y)$.

$$\mu(x,y)M(x,y) + \mu(x,y)N(x,y)\frac{\mathrm{d}y}{\mathrm{d}x} = 0$$

is exact iff

$$\frac{\mathrm{d}(\mu M)}{\mathrm{d}y} = \frac{\mathrm{d}(\mu N)}{\mathrm{d}x} \Longleftrightarrow M\mu_y - N\mu_x + (M_y - N_x)\mu = 0$$

If we want an intergrating factor that's a function of x alone (as a special case), we can solve:

$$\frac{\mathrm{d}\mu}{\mathrm{d}x} = \frac{M_y - N_x}{N}\mu$$

Second-order Linear ODEs

A linear second order ODE has the form

$$P(t)y'' + Q(t)y' + R(t)y = G(t)$$

Provided that $P(t) \neq 0$, it can be expressed as

$$y'' + p(t)y' + q(t)y = g(t)$$

Linear Independence

$$W(y_1, y_2) = \det\left(\begin{pmatrix} y_1 & y_2 \\ y_1{}' & y_2{}' \end{pmatrix}\right) = y_1 y_2{}' - y_1{}' y_2$$

If y_1, y_2 are differentiable on an open interval I,
- If $W(y_1,y_2)(t_0) \neq 0$ at some point $t_0 \in I$, then y_1, y_2 are

linearly independent on I.
- If y_1, y_2 are independent on I then $W(y_1,y_2)(t) = 0$ for all $t \in I$.

Abel's Theorem

Suppose that y_1, y_2 are two solns to the homogenous ODE,

$$y'' + p(t)y' + q(t)y = 0$$

where p, q are coontinuous on an open interval I. Then the Wronskian

$$W(y_1, y_2) = C\exp\left(-\int p(t)dt\right)$$

Method of Reduction of Order

If we know one solution, to the linear homogenous ODE, then we can plug this in to reduce the order $y(t) = v(t)y_1(t)$.

Linear Homogenous ODEs with Constant Coefficients

$$ay'' + by' + cy = 0$$

has the characteristic equation $ar^2 + br + c = 0$.

If the solutions are real r_1 and r_2, then

$$y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

If the solutions are complex r_1, r_2, then

$$\alpha = -\frac{b}{2a}, \quad \beta = \frac{\sqrt{4ac - b^2}}{2a}$$

$$y(x) = e^{\alpha x}(C_1\cos(\beta x) + C_2\sin(\beta x))$$

Linear Non-Homogenous ODEs with Constant Coefficients

$$ay'' + by' + cy = g(t)$$

Then the complentary equation is the homogenous version (where we make $g(t) = 0$). If the solution to the nonhomogenous version is Y_1, Y_2, then their difference is a solution of the complementary equation.

$$Y_1(t) - Y_2(t) = C_1 y_1(t) + C_2 y_2(t)$$

If we let Y_1 be an arbitrary solution y_t to the nonhomogenous version, and Y_2 be a particular solution, then the general solution can be expressed as

$y(t) = y_{c(t)} + y_{p(t)}$

Where $y_{c(t)} = C_1 y_1(t) + C_2 y_2(t)$ and is called the complementary function.

So to solve we:
- Find the general solution y_c of the homogenous equation
- Find any particular solution
- Add them to get the general solution

There are two main ways of finding a particular solution:
- Method of Underdetermined Coefficients
- Method of Variation of Parameters

Method of Underdetermined Coefficients

$$g(t) = Me^{kt}$$

- If k is not a root of the characteristic equation, try $y_p(t) = Ce^{kt}$
- If k is a root of the characteristic equation, try $y_p(t) = Cte^{kt}$
- If k is a repeated root of the characteristic equation, try $y_p(t) = Ct^2e^{kt}$

$$g(t) = M\cos(kt) + N\sin(kt)$$

- If $\pm ik$ are not roots of the characteristic equation, try $y_p = C\cos(kt) + D\sin(kt)$
- If $\pm ik$ are roots of the characteristic equation, try $y_p = t(C\cos(kt) + D\sin(kt))$

$$g(t) = a_n t^n + a_{n-1}t^{n-1} + \ldots + a_1 t + a_0$$

- If 0 is not a root of the characteristic equation, try $y_p(t) = b_n t^n + b_{n-1}t^{n-1} + \ldots b_1 t + b_0$
- If 0 is a root of the characteristic equation, try $y_p(t) = t(b_n t^n + b_{n-1}t^{n-1} + \ldots b_1 t + b_0)$
- If 0 is a repeated root of the characteristic equation, try $y_p(t) = t^2(b_n t^n + b_{n-1}t^{n-1} + \ldots b_1 t + b_0)$

Method of Variation of Parameters

The general form is this primeval chaos-monster:

$$y_p(t) = -y_1(t)\int\left(\frac{y_2 g(t)}{aW(y_1, y_2)}\right)dt + y_2(t)\int\left(\frac{y_1(t)g(t)}{aW(y_1, y_2)}\right)dt$$

Reducible ODEs

A general 2nd-order ODE has the form,

$$F(y'', y', y, x) = 0$$

An ODE,

$$F(y'', y', x) = 0$$

Can be solved using the change of variable $v(x) = y'$.

Euler Equations

An Euler Equation has the form,

$$ax^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + bx\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0$$

Where $a, b, c \in \mathbb{R}$ and $a \neq 0$.

The leading coefficient vanishes at $x = 0$, so we if we restrict our attention to $x > 0$, we can use the dependent variable $t = \ln x, x > 0$, which gives us the equivilent form

$$a\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + (b-a)\frac{\mathrm{d}y}{\mathrm{d}t} + cy = 0$$

Which we can then solve like a Linear Homogenous ODE with Constant Coefficients (characteristic equation $ar^2 + (b-a)r + c = 0$).

Series Solutions for Linear Homogenous ODEs

We can use a power series to construct a fundamental set of solutions,

$$\sum_{n=0}^{\infty} a_n(x-b)^n$$

The **radius of convergence**,

$$\rho = \lim_{n\to\infty}\left|\frac{a_n}{a_n} + 1\right| \geq 0$$

means that,

$$\sum_{n=0}^{\infty} a_n(x-b)^n \text{ converges absolutely for } |x-b| < \rho$$

$$\sum_{n=0}^{\infty} a_n(x-b)^n \text{ diverges for } |x-b| > \rho$$

$$\sum_{n=0}^{\infty} a_n(x-b)^n \text{ either converges or diverges for } |x-b| = \rho$$

Power series are unique.

Consider the homogenous ODE

$P(x)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + Q(x)\frac{\mathrm{d}y}{\mathrm{d}x} + R(x)y(x) = 0$

Or if $p = \frac{Q(x)}{P(x)}$ and $q = \frac{R(X)}{P(X)}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + p(x)\frac{\mathrm{d}y}{\mathrm{d}x} + q(x)y(x)$

The point $x = a$ is an **ordinary point** of the equation if $p(x)$ and $q(x)$ are both analytic at $x = a$; otherwise $x = a$ is a **singular point**.

If $P(x)$, $Q(x)$, $R(x)$ are polynomials with no common factors, then $x = a$ is a singular point iff $P(a) = 0$, else an ordinary point.

If $x = b$ is an ordinary point of the ODE, then the general solution can be expressed as

$$y(x) = \sum_{n=0}^{\infty} a_n(x-b)^n = C_0y_1(x) + C_1y_2(x)$$

Further, y_1 and y_2 are linearly independent, the radius of convergence for each of them is at least as large as the minimum of the radiuses of convergence of the power series for p qnd q. The coefficients can be determined by substitution into the equation.